Engineering Failure Analysis 134 (2022) 106035

Contents lists available at ScienceDirect

Engineering Failure Analysis

journal homepage: www.elsevier.com/locate/engfailanal

Experiments and numerical analyses on splitting fracture of wire under multi-pass drawing

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ARTICLE INFO

Keywords:
Multi-pass drawing
Splitting fracture
Damage behavior
Drawing parameters
Numerical simulation

ABSTRACT

The splitting phenomenon during drawing will cause the quality decline of wire. A clear understanding of formation mechanism of splitting can help to promote the development of engineering drawing. Therefore, the splitting fracture and microstructure are analyzed by SEM. The experimental results indicate that the main crack run through the core and surface of both sides of wire. Furthermore, multi-pass drawing of different material properties and die angles is simulated by using FEM with the developed damage modeling. The results show that after each drawing pass, the damage and section stress are formed near the center and surface of wire, respectively. Meanwhile, the damage will be enhanced with the increase of die angle, drawing strain and material properties. Among these influencing factors, the die angle has little effect on the section stress, but the increase of the other two factors will lead to the gradual increase of section stress. Moreover, wire core becomes the source of crack due to its larger damage, and then crack propagates to the surface of wire under the action of stress, thus forming splitting fracture.

1. Introduction

Steel wires, with a desirable combination of high strength, good ductility and wear resistance, are widely used in civil, mining and other related engineering fields[1,2]. The wires are typically produced by cold drawing technique, one of the most frequently used methods in wire manufacturing industry[3]. The integrity of wire needs to be ensured in the wire drawing process. Once the wire breaks, it will not only make the product unqualified, but also cause the production line to stagnate [4]. Therefore, the wire broken in drawing is still a hotspot concerned by engineers and technicians, among which splitting fracture [5] often occurs in multi-pass drawing. This kind of fracture does not break directly at some point of wire, but breaks into two (or more) parts continuously and uniformly on the central axis of wire, which is destructive to a certain extent. In general, it is necessary to clarify the influencing factors of wire breakage caused by splitting in order to improve the production process and solve the problem of wire breakage [6].

There are various types of fracture in the drawing process, and these fractures are closely related to factors like drawing conditions, and fracture behavior etc.[7,8]. Hence, the causes of different fractures should be analyzed in detail. From the experimental point of view, Browning et al.[9] explored the effect of temperature on wire splitting, and the results showed that wire splitting could be avoided when the drawing temperature stayed high. Mukhopadhyay et al.[5] studied splitting fracture via microscopic examination, fractography and mechanical tests, and the results indicated that there was a certain relationship between splitting fracture and

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https://doi.org/10.1016/j.engfailanal.2022.106035
Received 18 September 2021; Received in revised form 3 January 2022; Accepted 4 January 2022
Available online 8 January 2022
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Nomenclature

HSM	high-strength material
LSM	low-strength material
d_{in}	inlet wire diameter
d_{out}	outlet wire diameter
α	die half-angle
σ_{eq}	effective stresses of porous aggregate
σ_m	mean stresses of porous aggregate
σ_s	yield (flow) stress of material
q_1,q_2,q_3	material correction coefficient
f^*	void volume fraction
ε_{eq}^p	corresponding equivalent plastic strain
S_n	standard deviation
ε_n	mean nucleating strain
f_n	void volume fraction of void nucleating particles
ε	drawing strain
d_0	the diameter of the initial wire
d	the wire diameter after drawing
μ	coefficient of friction
D_{max}	the maximum value of the void volume fraction

hydrogen embrittlement. In the course of experiment, scholars found that the pearlite phase had fibrous deformation in the longitudinal direction of drawing [10,11], which made the longitudinal deformation weaker than the transverse one [12]. With the progress of drawing, the crack could slowly approach the cold drawing direction of the wire rod [13,14], and the increase of inclusions would induce microcracks [15–17]. According to microstructural observation, splitting tended to be a kind of brittle fracture, and brittle grain boundary fracture occurred in most cases [18]. For instance, Gaal et al.[19] revealed the mechanism of fracture nucleation along the originally longitudinal grain boundaries by means of free-end torsion. Ocenasek et al.[20] regarded splitting as a kind of intergranular crack and successfully predicted the fiber texture evolution by establishing a micro grain structural mode. Ripoll et al.[21] concluded by simulation that there was a large tensile stress concentration near the grain boundary, and these stresses might be the cause of splitting. Zhu et al.[22] investigated the mechanism of microstructural evolution through micro-observation and LS-dyna finite element analysis software, which helped to better understand the relationship between stress distribution and microstructural evolution in the process of multi-pass cold drawing. Besides, due to the non-uniformity of drawing process, residual stress could be produced in the drawing process [23], affecting the mechanical properties of final product, such as fracture tendency and life [24]. The most effective way to obtain residual stress is to realize the visualization of stress distribution through numerical simulation. Weygand et al.[25] considered that residual stress was one of the factors affecting splitting fracture, so the effects of back stress and die friction on residual stress were explained by introducing Chaboche-type plasticity model into finite element method (FEM). Ripoll et al. [26] designed a stepped mold, and found that residual stress of all components was significantly reduced after simulating the new mold by ABAQUS. The calculation results further demonstrated that the bending of wire drawing at high temperature also helped to reduce residual stress. In view of the harmfulness of residual stress, Toribio et al.[27] proposed a novel die design method, which was using two continuous die angles (i.e., varying die angle) to reduce residual stress and strain state of cold-drawn wire to improve the hydrogen embrittlement resistance of prestressed steel wire. Baumann et al.[28] established dies with different geometric shapes by FEM to reduce the residual stress distribution for the enhancement of crack resistance of wire in the drawing process.

According to the previous studies, although scholars have done a lot of work on the analysis of wire drawing, the understanding for the formation mechanism of splitting is still missing, which is urgent to be solved in engineering. Besides, wire breakage caused by splitting is still rarely studied from the aspects of material properties and drawing parameters. The main reason is that the effects of material properties and drawing parameters on the damage and splitting of wires can't be quantitatively obtained from experimental research. Therefore, the splitting phenomenon in the drawing process and the microstructure near the fracture surface are qualitatively analyzed by scanning electron microscope (SEM). Based on the experimental research, multi-pass drawing simulation is carried out for different material properties and different die angles via finite element software ABAQUS. The simulation results can not only help to understand the formation mechanism of splitting fracture, but also obtain the damage behavior of different materials at different die angles, so as to better provide theoretical support for the design of die angle of drawing materials.

2. Experimental study on splitting fracture of wire drawing

In this section, the splitting fracture of wire drawing is analyzed by SEM, and the crack propagation mechanism is studied through macro phenomenon and microstructure. Meanwhile, in order to explore the influence of material properties on drawing, materials with different strengths are specially selected. Mechanical properties of the selected materials are tested by testing machine to provide different mechanical parameters for later numerical simulation.

2.1. Materials and mechanical property for the experiment

The damage of different wires in the drawing process is different, which makes the corresponding drawing parameters different. In this section, pearlitic steel wire and austenitic stainless steel wire are selected for tensile test to obtain the main mechanical parameters. Tensile strength of wires is determined by tensile tests at room temperature using a INSTRON 2344 type universal testing machine, operating at a constant speed of 2 mm/min. Tensile tests are performed according to Chinese National Standard GB/T228-2002. The stress-strain curves of these two materials are shown in Fig. 1.

It can be clearly seen from Fig. 1 that the strength of pearlite steel wire is obviously higher than that of stainless steel wire. Therefore, in this paper, the selected pearlitic wire represents high-strength material (HSM), and the selected stainless steel wire represents low-strength material (LSM). Specifically, the mechanical parameters of two materials are shown in Table 1.

2.2. Observation on drawing fracture and microstructure

Fracture often occurs in the process of wire drawing, and different external factors will generate different fracture patterns[29]. Based on previous drawing test studies, the drawing fracture types and related influencing factors are listed in Table 2.

It can be known from Table 2 that different fracture types will be generated due to different factors or the superposition of factors. In experiment, we also found that the splitting fracture will appear in the drawing process. In order to have a better understanding of this fracture type, the morphology of splitting fracture, the microstructure near the fracture and the secondary cracks are observed by Sirion-400 field emission scanning electron microscope (SEM) in this section. Fig. 2 shows the fracture morphology and crack propagation of different materials during drawing. The corresponding drawing parameters are mainly as follows: die half-angle is 10°, the cross-section reduction ratios of each pass are about 20%, and the drawing strain is above 2.

Fig. 2 (A) and (B) are some morphological features of pearlite steel wire after drawing fracture. In the figures, a large major crack is formed through the core and surface of wire, and the width of crack in the core is larger than that on the surface, which tells that the major crack appears in the core of wire. With the progress of drawing, the core crack propagates to the surface of wire, thus forming major crack. Meanwhile, the surface of wire produces some certain secondary cracks that are mostly perpendicular or approximately perpendicular to major crack. There are also some tearing areas at the opening position of secondary cracks (see Fig. 2 (A)), and the width of the tearing area on the surface of wire is larger than that inside the wire, indicating that the secondary crack mainly arises from the surface of wire and then extends to the core. The microstructure in Fig. 2 (b) suggests that there are many micro voids in the wire.

Fig. 2 (C) and (D) are splitting crack propagation generated by stainless steel wire as the drawing strain continues to increase. An obvious longitudinal crack in the middle of the wire surface can be seen from Fig. 2 (C). In order to better observe the crack propagation and distribution in the cross section of wire, a small section of wire with longitudinal crack was cut off for sample preparation. After grinding and polishing, the crack distribution in the cross section of the sample was observed by SEM, as shown in Fig. (d). Through the observation of magnification, it is obvious that a crack first appears in the center of wire. Except for this crack, there are no

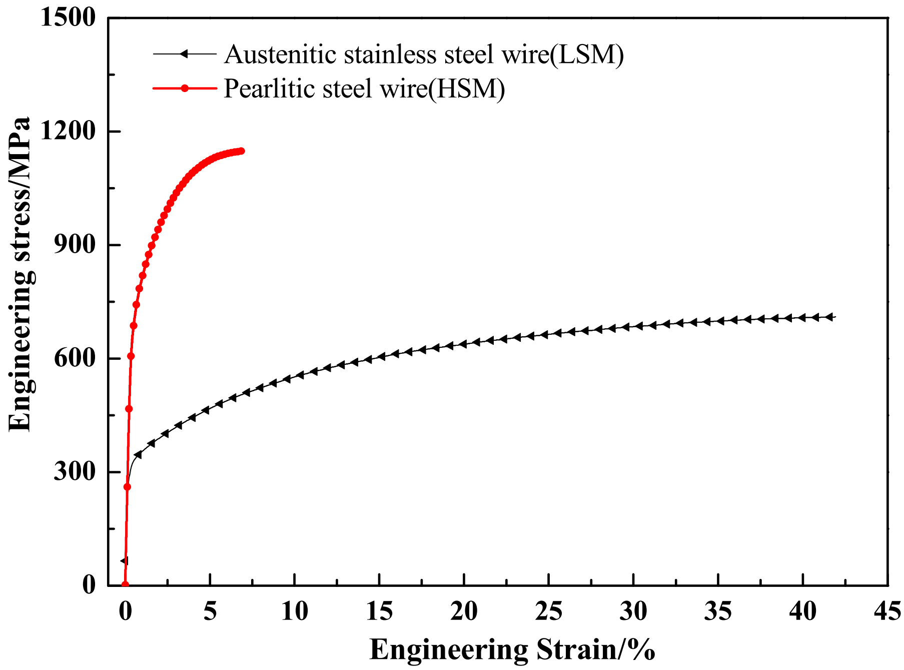
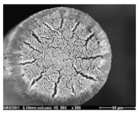
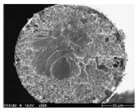
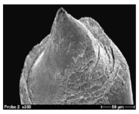
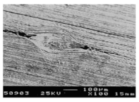
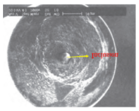
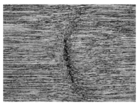


Fig. 1. Engineering stress-strain curves of materials.

Table 1
The related mechanical parameters.

Materials	Pearlitic steel wire	Stainless steel wire
Young's modulus (GPa)	200.091	194.247
Poisson's ratio	0.3	0.28
Density (t/mm^3)	7.8E-9	7.72E-9
Yield stress (MPa)	466.212	215.614
Tensile strength (MPa)	1148.138	709.868

Table 2
The type of fracture and analysis of related influencing factors during wire drawing[5–7,30–32]

Fracture phenomenon	The type of drawing fracture	The analysis of related influencing factors
	Ductile fracture	Drawing stress too high; Drawing temperature too high; Back pull stress too high; Friction within the die too high
	Brittle fracture	Additional bending stresses at the drawing drum; Surface defect
	Cup cone fracture	Drawing parameters too large; Inadequate lubrication; Back pull stress too high
	Inclusion fracture	Type, size and distribution of inclusion; Influence of external hard inclusions
	Pen-tip shaped fracture	Central inclusion
	Crows feet fracture	Disadvantageous die geometry; Insufficient lubrication; An inclined wire feeding into the die
	Splitting fracture	Hydrogen embrittlement; Unreasonable drawing parameters; Too high strengthening; Deformation temperature too low; Strong radial gradients of residual stresses (tangential stresses)

other cracks in the cross section of wire. The main reason is that steel wire accumulates large stress and damage with the increase of drawing strain, and when they reach the critical value at a certain time, microcracks will appear preferentially in this position. The generation of microcracks will lead to stress release, so that microcracks will not be formed immediately in other locations. However, with the continuous increase of drawing strain, the width of the original microcrack will keep growing, thus forming an obvious macroscopic splitting. Some secondary microcracks will also appear in other directions. As a result, it will lead to the failure of wire drawing.

Through the drawing fracture and crack propagation of the two materials in Fig. 2, it can be known that when the die angle and drawing strain are large, the splitting phenomenon will appear. Especially when the strength of steel wire is high, the crack propagation is more serious, and the splitting is more obvious.

3. Numerical modelling of wire drawing

3.1. Simplified physical model and basic assumptions

The above experimental results and analysis verify that splitting fracture will occur during wire drawing. The microstructure of wire shows that micro voids will finally develop to be a crack after growing and passing through among each other during drawing. In this process, the wire is damaged in varying degrees, so the introduction of damage evolution into wire drawing is conducive to figuring out the formation mechanism of fracture. At the same time, the change of material and drawing parameters will bring forth different damage values, and the distribution of damage also directly affects the morphology of fracture. Therefore, material properties

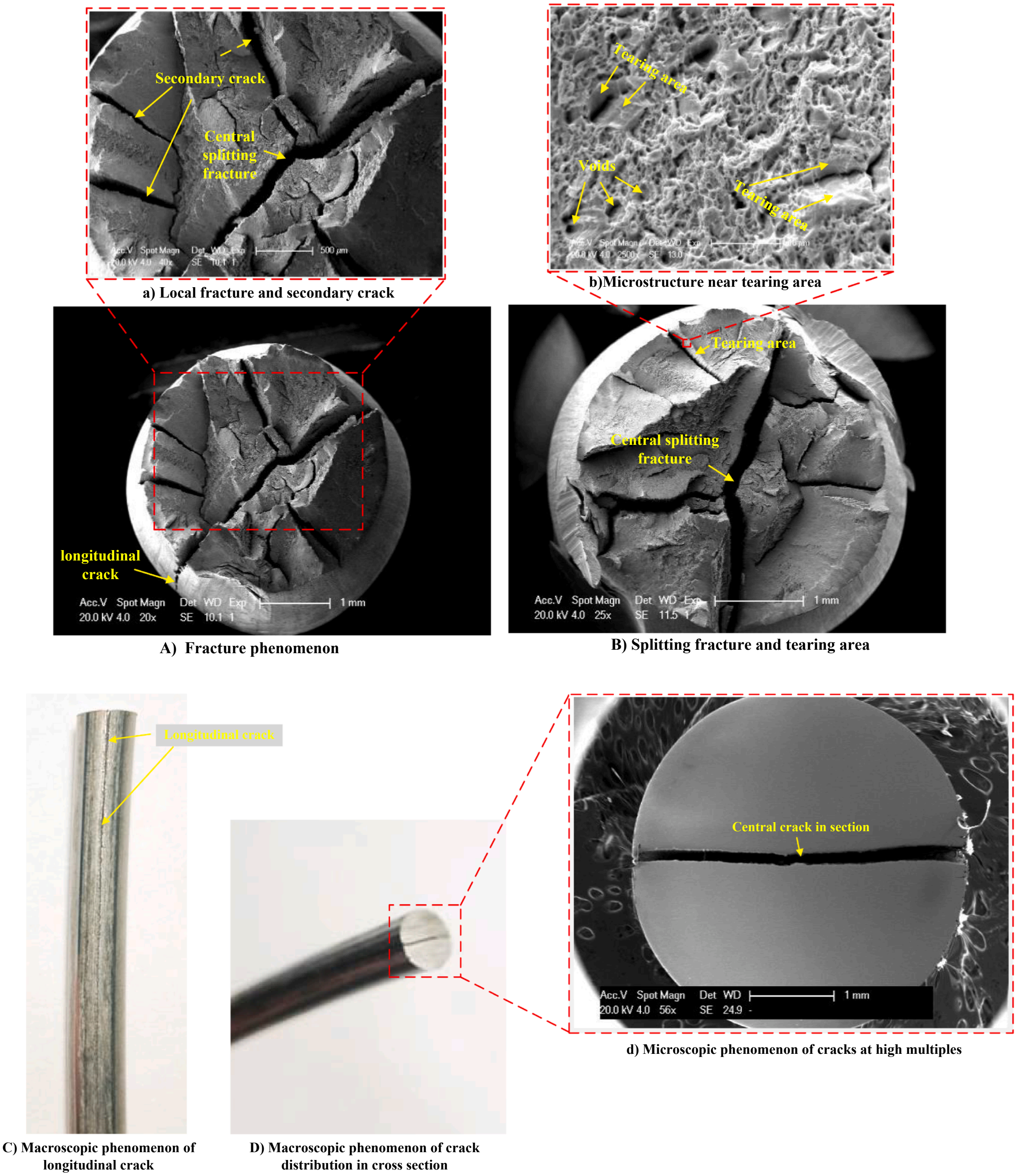


Fig. 2. Drawing fracture and microstructure phenomenon of wire.

and die angles are taken into account during the simulation of wire drawing. The detailed schematic diagram of the simplified model of wire drawing process is shown in Fig. 3.

In order to effectively simulate the wire drawing damage process, it can be simplified as follows:

1) The die is regarded as a rigid body, and the friction coefficient between the die and the deformable body does not change in the drawing process.

2) It is assumed that the stage after the ultimate strength of material is regarded as a hardening process that increases linearly with the increase of plasticity. The reason for this assumption is that when the material is under uniaxial tension, it occurs fracture with lower strain. However, during wire drawing, wire rod can reach a larger strain value, and the yield strength of the material after drawing has been greatly improved.

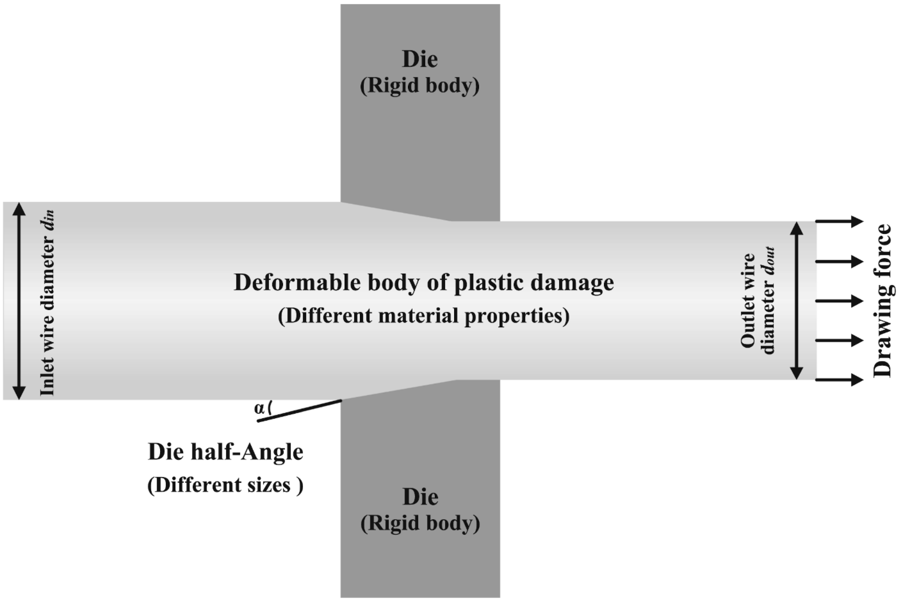


Fig. 3. Schematic diagram of the simplified model of wire drawing.

3) Deformable body is considered to be homogeneous and isotropic, and also regarded as a plastic damage material with micro voids. That is because there are a large number of micro voids in the microstructure of wire, and the wire drawing is a plastic flow forming process.

3.2. *Plastic damage evolution law for large deformable body*

The Gurson Tvergaard Needleman (GTN) is a phenomenological physical model based on mesoscopic damage parameters which is linked with void volume fraction, and the volume change can be described by the growth of void. Therefore, this model can better reflect the microstructure of material than the other damage models. Meanwhile, the GTN model has an advantage of being simple for implementation in standard FE codes, and has been widely used in ductile damage community[33–35]. The following is a detailed introduction about the model.

The plastic potential of ductile failure predicted by GTN model can be expressed as[36,37]:

$$\Phi = \left(\frac{\sigma_{eq}}{\sigma_s}\right)^2 + 2f^* q_1 \cosh\left(q_2 \frac{3\sigma_m}{2\sigma_s}\right) - \left(1 + q_3 f^{*2}\right) = 0 \tag{1}$$

where σ_{eq} and σ_m represent effective and mean stresses of porous aggregate, σ_s is the yield (flow) stress of material, q_1 , q_2 and q_3 are the material correction coefficient, f^* is the void volume fraction, and the expression is as follows:

$$f^* = \begin{cases} f, f < f_c \\ f + \dfrac{f_u^* - f_c}{f_F - f_c}(f - f_c), f \geq f_c \end{cases} \tag{2}$$

where f_{c} is the void volume fraction when the voids begin to converge, and f_{F} is the marginal void volume fraction while fracture occurs. The material point loses its loading bearing capacity fully when $f = f_{\mathrm{F}}$ or $f^* = f_{\mathrm{u}}^* = 1/\mathrm{q}_1 . f_{\mathrm{u}}^*$ means 100% squeezing of yield surface compared to the unchanged yield surface while $f = 0$.

In the process of damage accumulation, the law of damage evolution can be described as [38]:

$$\dot{f} = \dot{f}_{growth} + \dot{f}_{nucleation} \tag{3}$$

where $\dot{f}_{\mathrm{growth}}$ represents the rate of volume change caused by the growth of microvoids, $\dot{f}_{\mathrm{nucleation}}$ represents the rate of volume change caused by the nucleation of microvoids.

The growth of voids depends on the change of macroscopic shaping volume, and the expression is as follows:

$$\dot{f}_{growth} = (1 - f)\dot{\varepsilon}_{kk}^p \tag{4}$$

For the nucleation of voids, strain-dominated type is used in the calculation, which can be expressed as follows:

$$\dot{f}_{nucleation} = A \bullet \dot{\varepsilon}^{p}_{eq} \tag{5}$$

$$A = \frac{f_n}{S_n\sqrt{2\pi}}\exp\left\{-\frac{1}{2}\left(\frac{\varepsilon^p_{eq}-\varepsilon_n}{S_n}\right)^2\right\} \tag{6}$$

where A is the strength function of void nucleation, ε^p_{eq} is the corresponding equivalent plastic strain, ε_n is the mean nucleating strain, f_n is the void volume fraction of void nucleating particles, and S_n is the standard deviation.

There are many researches on the parameter selection of GTN model. The previous studies never stop selecting the best value through the combination of experiments and numerical simulation, and they constantly verify the rationality of the parameters via numerical simulation. According to the relevant research results[37,39–42], the model parameters are shown in Table 3 below.

3.3. Model verification and numerical scheme

The experiment can only qualitatively explain that the shape of drawing fracture is related to material damage, but it is difficult to quantitatively determine and analyze the fracture behavior in drawing. In order to explain the stress and damage distribution of different materials at different die angles and splitting fracture in the process of wire drawing, multi-pass drawing analysis is carried out by finite element method.

In calculation and analysis, the model is established by control variable method. The variables are material properties under two strengths and drawing parameters under four different drawing angles. The material properties apply the parameters mentioned in Section 2, and die half-angle(α) is set as 6°, 8°, 10°, 12°. Numerical models of steel wire (5.5 mm in diameter) are drawn successively for five passes to a diameter of 3.15 mm. The cross-section reduction ratios of each pass are maintained at 20%±0.2%[43], and the total reduction in area is 67.2%. The drawing strain ($\varepsilon = 2 * ln(d_0/d)$[44], where d_0 is the diameter of the initial wire and d is the diameter after drawing) of each pass is 0.223, 0.446, 0.667, 0.893 and 1.115, respectively. The drawing speed and friction coefficient (μ) are set as 5 m/s and 0.05[45], respectively.

In this study, a three-dimensional drawing model was established by ABAQUS, and explicit analysis is used. Since die deformation is ignored during drawing, the rigid material of die is selected to define the mechanical parameters. The total simulation time is 0.048 s, and the 8-node linear structured hexahedral element of type C3D8R is used in the model. The global model construction and mesh generation are shown in Fig. 4. The boundary conditions in the simulation are included in Fig. 4 (A). The right end of wire is mainly set as the drawing surface, and the operation of wire drawing is carried out by means of displacement control loading. The contact type between the surface of steel wire and the inner surface of die is set to surface-to-surface contact. The displacement in three directions (U_1, U_2, U_3) and the rotation in the three directions (UR_1, UR_2, UR_3) are all constrained to fix the die. During this process, a reference point should be first set on the die, and then the reference point and the die are constrained by the type of rigid body, so that the constraint of the die can be realized directly by constraining the reference point.

In order to shorten the calculation time, mass scaling factor is used in this simulation, and the value is set to 100. The energy variation of calculation results is shown in Fig. 5 to verify the rationality of mass scaling factor. Fig. 5 further demonstrates that the ratio of kinetic energy to internal energy is less than 5%, indicating that the data calculated by the model is available.

The above material model and corresponding mechanical characteristic parameters are used for numerical simulation calculation to verify the correctness of the model. Fig. 6 is the section stress distribution of two materials after the second pass (α = 6°) and the morphology of splitting in the experiment. Fig. 6 (A) and (B) show the symmetrical section shear stress in the cross section of steel wire. Except for the difference in magnitude, the distribution of stress produced by the two materials is basically similar. Through the further superposition and conversion of these section shear stresses, it can be known that the axisymmetric compression stress will be distributed on the near surface of steel wire after drawing, which is consistent with the splitting fracture results in Fig. 6 (C). The above comparison proves that these models are suitable for simulation analysis of wire drawing.

4. Numerical simulation of multi-pass drawing

4.1. Damage distribution after each pass drawing

In this research, the drawing damage is mainly represented by the void volume fraction (f^*) of the GTN model, so the damage mentioned below all represent the void volume fraction. The damage distributions of high-strength and low-strength materials after each pass are shown in Fig. 7 and Fig. 8, respectively, where the initial location of drawing damage is in the internal position of wire. With the increase of drawing pass, the diameter of wire slowly decreases, and the corresponding damage gradually concentrates on the core of wire. The main damage area is finally shown in the central position.

Table 3
The related parameters of GTN model.

Materials	q_1	q_2	q_3	S_n	f_n	ε_n	f_c	f_F
Pearlitic steel	1.5	1	2.25	0.1	0.04	0.3	0.15	0.25
Stainless steel	1.5	1	2.25	0.09	0.032	0.54	0.11	0.156

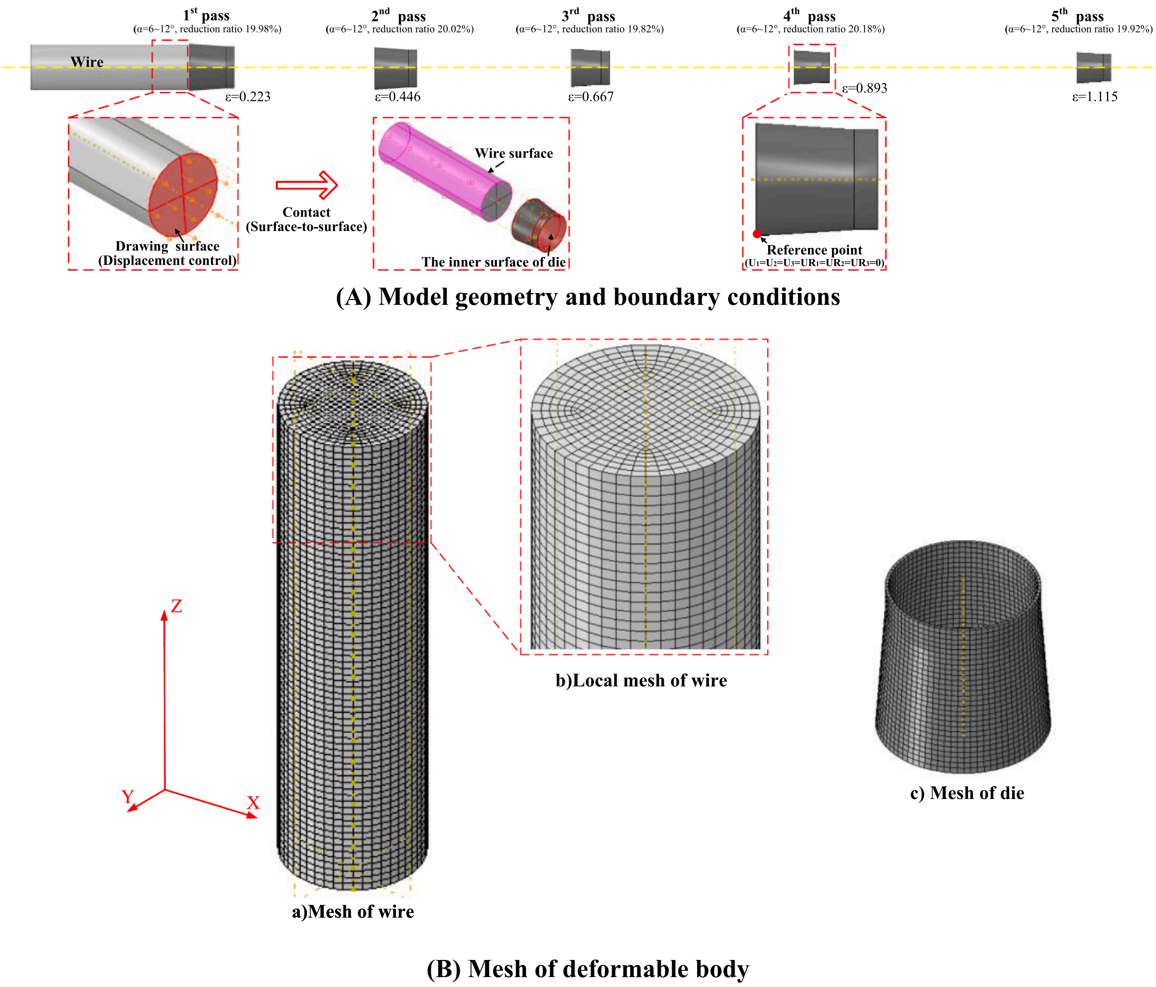


Fig. 4. Global model construction and mesh generation.

In order to better compare the drawing damage caused by the two materials at different angles, the range of damage is set to be the same value. It can be found from the figure that the die angle and drawing strain are the key factors affecting the damage when the materials are the same. At the same angle, the increase of drawing strain is the main factor leading to the central damage, and with the increase of die angle, the damage under the same drawing strain will also be aggravated. Fig. 7 and Fig. 8 revealed that materials also have a great influence on the drawing damage. Compared with high-strength material, the damage range of low-strength material decreases after multi-pass drawing, and the damage degree is also reduced. Meanwhile, there is no obvious damage in low-strength material after the first two drawing passes at lower angles, and the damage mainly comes from the accumulation of the last several drawing passes. With the increase of angle, low-strength material gradually appears some obvious damage areas in the first two passes.

4.2. Stress evaluation of wire in multi-pass drawing

Fig. 9 and Fig. 10 show the cross-section stress distribution of two kinds of strength materials under different drawing strains and die half-angles. Through conversion and synthesis for the shear stress at the main stress concentration positions in the diagram, it can be known that wire is mainly affected by symmetrical compression stress in its cross section after each drawing strain. The stress is mainly concentrated on the surface of wire and extends gradually from the outside to the inside. The stress range increases with the increase of drawing strain under the same drawing parameters. When drawing strain is the same, the stress range at each angle is basically similar, but stress distribution range is different. The stress at lower die angle has a larger coverage inside the wire. With the increase of angle, the stress range inside the wire gradually decreases, and stress is progressively concentrated on the near surface of wire. The section stress of material with different strength varies greatly. Concretely, through the comparison of section stress at the same angle and drawing strain in Figs. 9 and 10, the stress range of high-strength material is obviously higher than that of low-strength material.

Based on the above analysis for damage and stress distribution during wire drawing, the damage will increase with the increase of

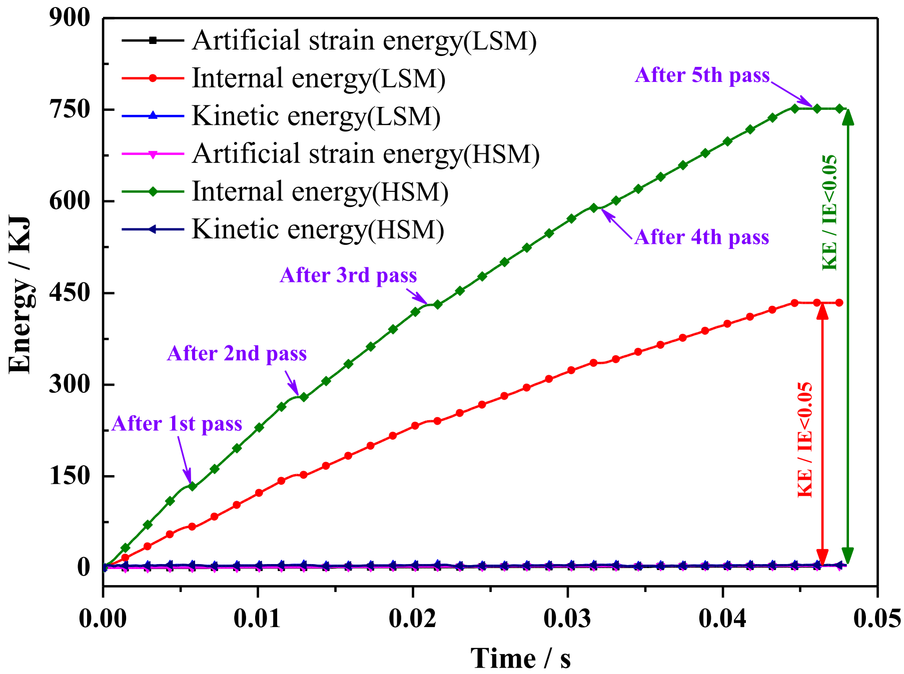


Fig. 5. Energy change in numerical calculations.

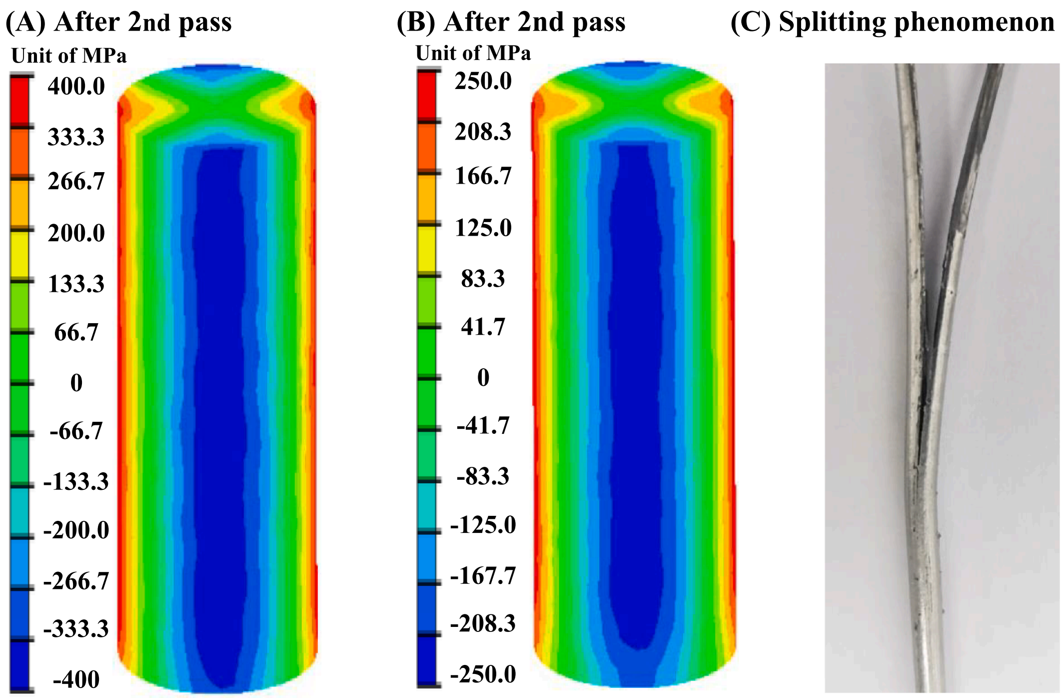


Fig. 6. (A) The section stress(S_{XY}) distribution of pearlitic steel wire($\alpha = 6^{\circ}$); (B) The section stress(S_{XY}) distribution of stainless steel wire($\alpha = 6^{\circ}$); (C)The splitting phenomenon of experiment.

die angle. At the same time, drawing damage is mostly concentrated in the core position of wire with the progress of drawing, and the initial crack will also occur at these positions. Furthermore, there are large symmetrical compression stresses on the surface of wire after each drawing pass based on the characteristic of stress distribution of wire section. Therefore, these compressive stresses will cause the initial crack to expand outward from wire core, and splitting fracture will then be generated due to the superposition of the

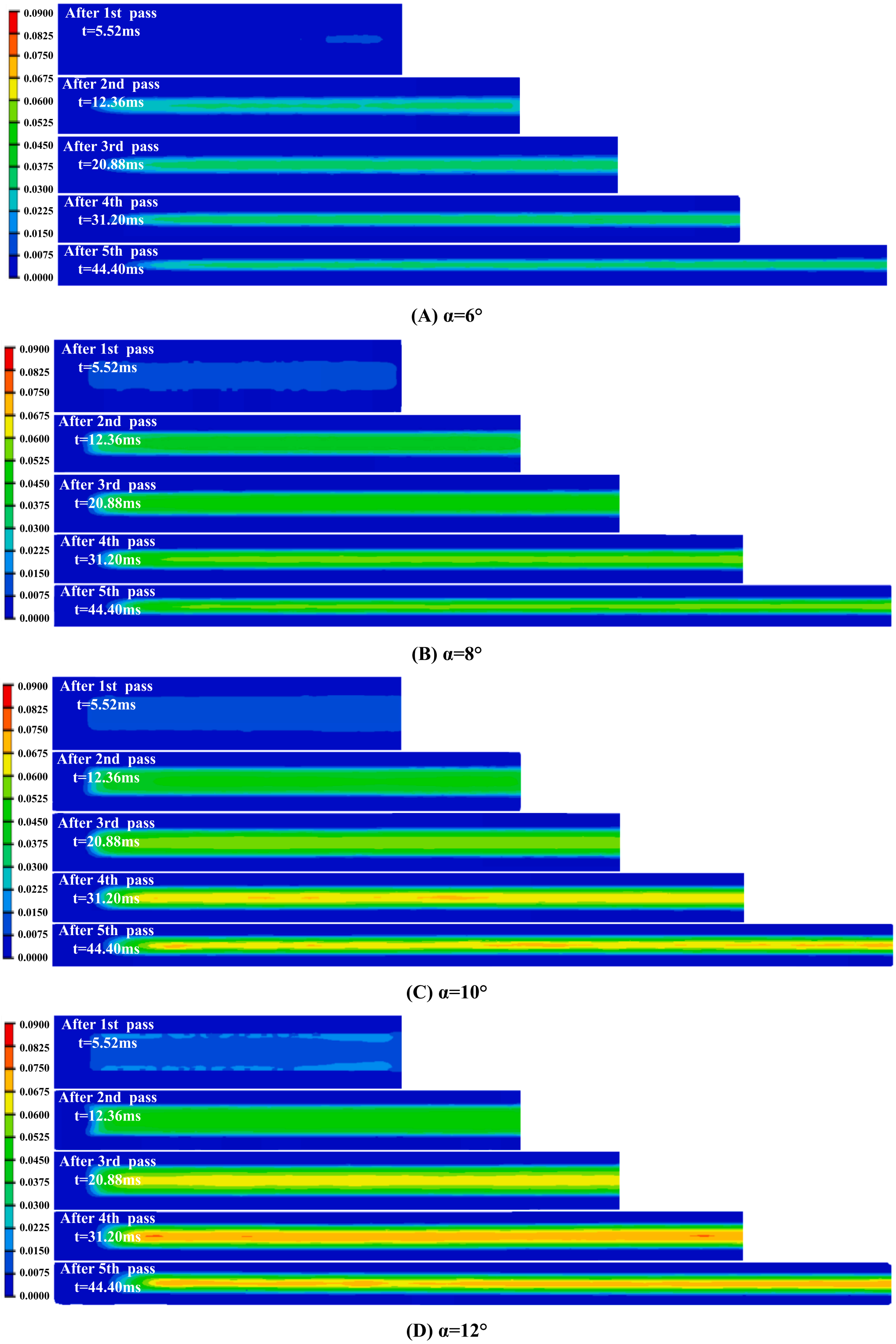


Fig. 7. Damage nephogram of high-strength material after each pass drawing.

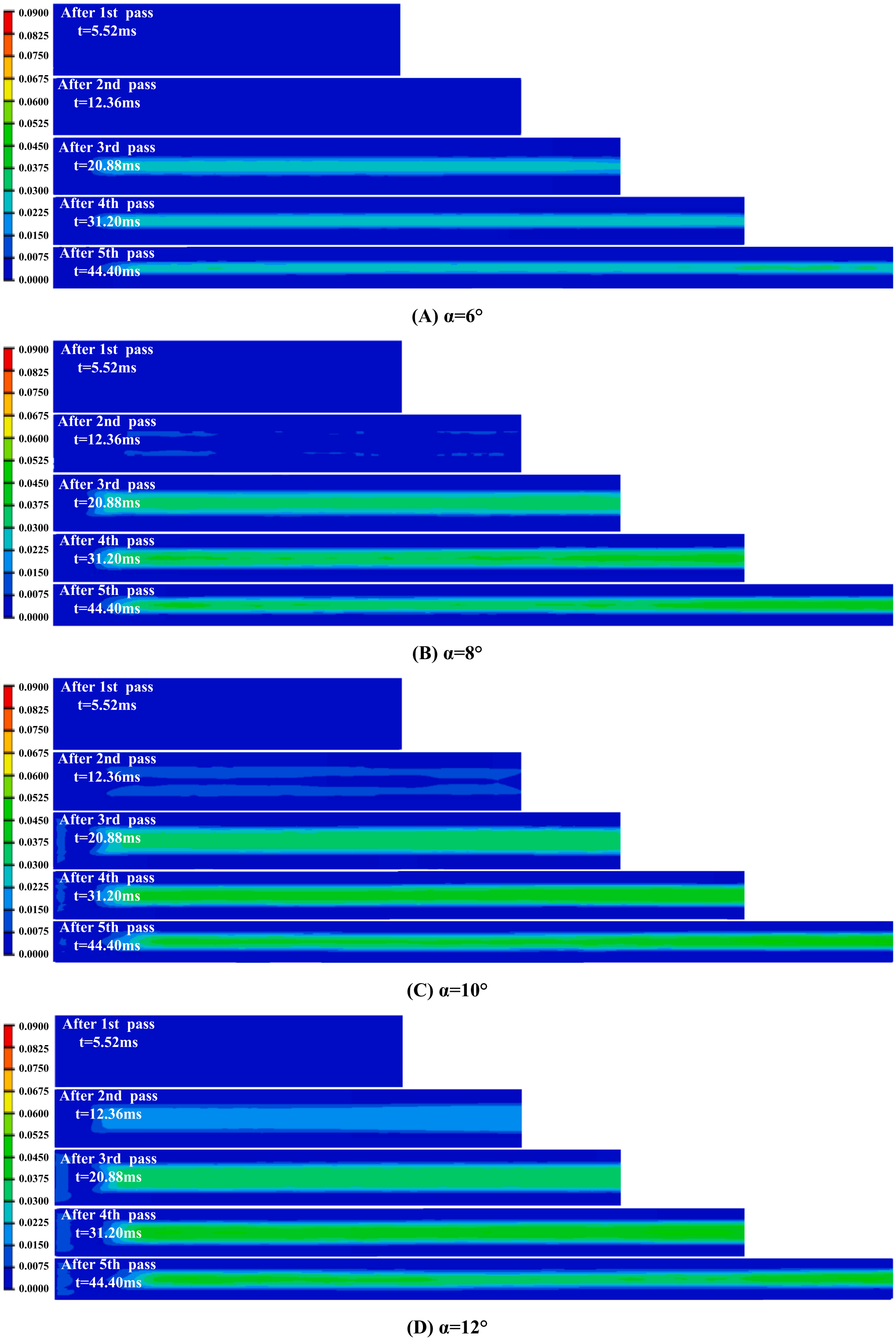


Fig. 8. Damage nephogram of low-strength material after each pass drawing.

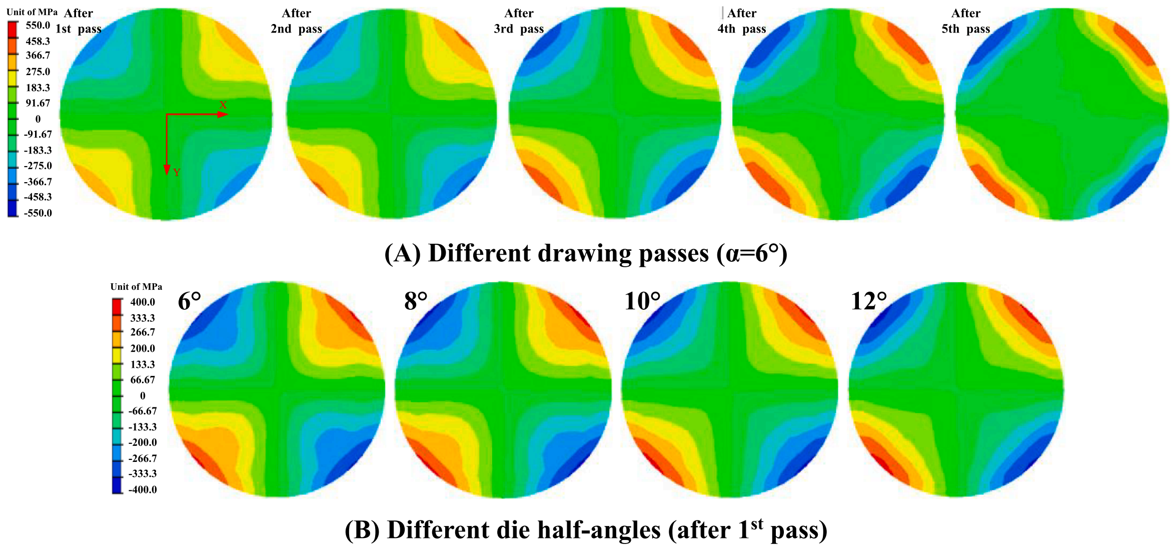


Fig. 9. Section stress nephogram of high-strength material.

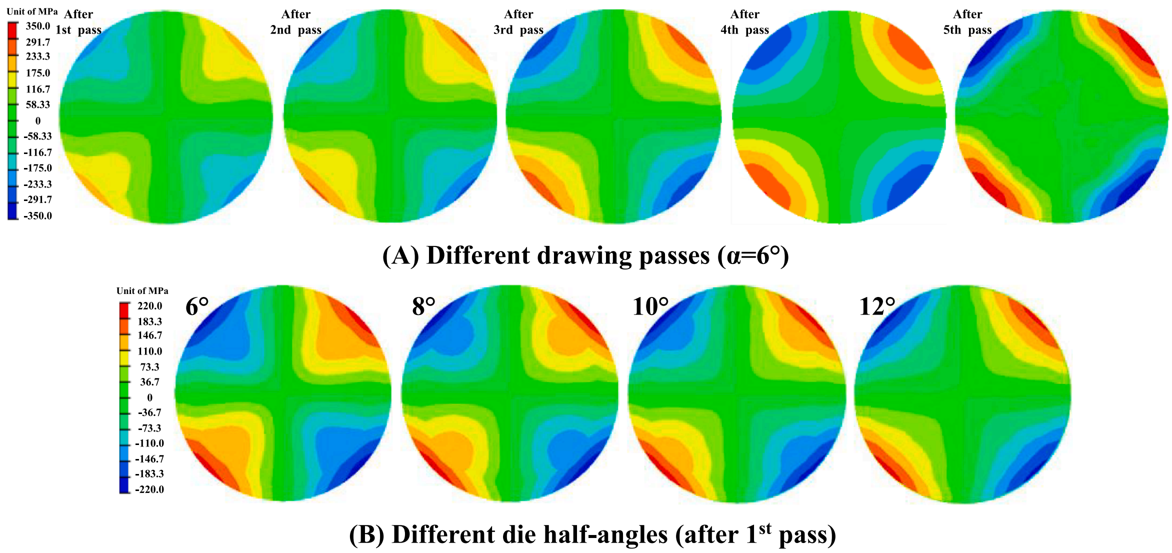


Fig. 10. Section stress nephogram of low-strength material.

core damage and the section stress.

5. Analyses on splitting of wire drawing with different material properties and die angles

5.1. Effect of material properties on splitting

The numerical nephograms in the previous section show the distribution trend of damage and stress in the multi-pass drawing process. But it is difficult to obtain specific values due to the wide range of values in numerical nephogram. In this section, the corresponding values of each drawing strain are obtained from simulation results, and also the damage and section stress of materials with different strength are analyzed. Fig. 11 lists the maximum damage (D_{max}) of materials with different strength at each half-die angle during drawing. The maximum damage (D_{max}) mentioned is the maximum value of the void volume fraction in the GTN model. Fig. 12 enumerates the maximum section stress under each drawing pass in terms of half-die angle $\alpha = 6^{\circ}$. The data of figure are fitted by the least square method, which is also applicable to the following fitting.

As shown in Fig. 11, when $\alpha = 6^{\circ} \sim 12^{\circ}$, the maximum damage value increases with the increase of drawing strain in a logarithmic trend. In the whole drawing process, it can be clearly found that there are obvious differences in drawing damage of materials with

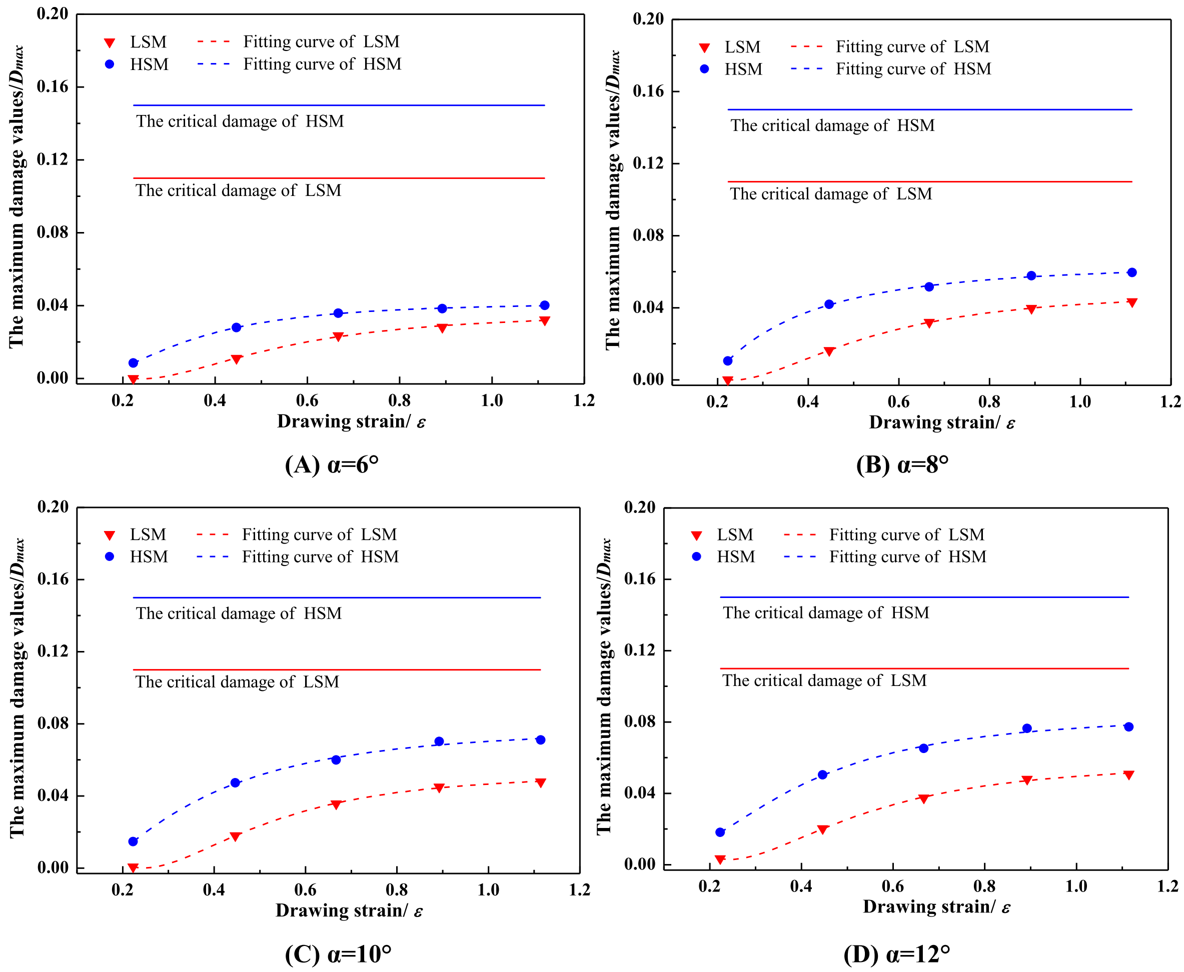


Fig. 11. The drawing damage at different die half-angles.

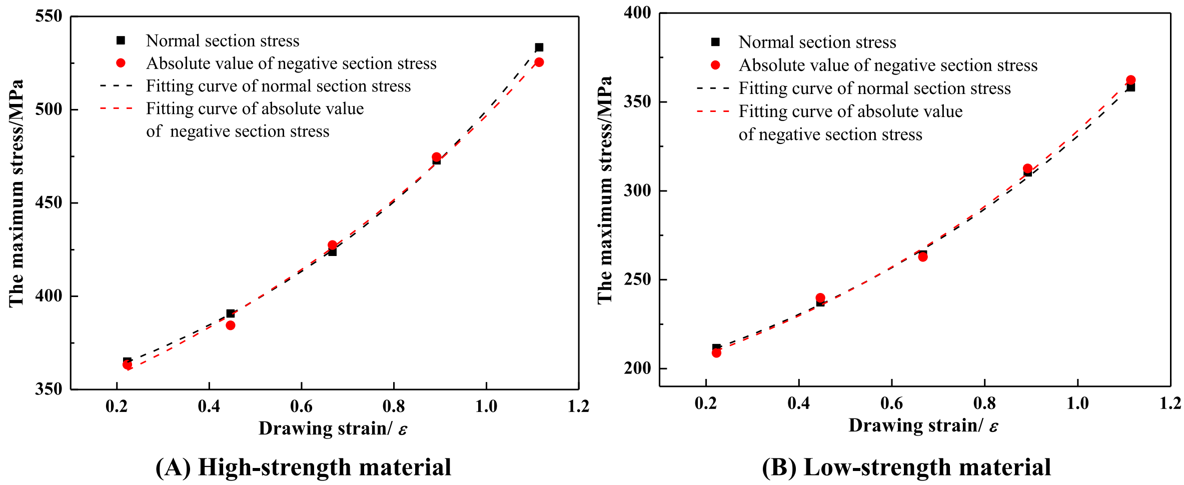


Fig. 12. Section stress tendency of different drawing passes (α = 6°).

different strength. The maximum damage of high-strength material is greater than that of low-strength material at all angles and drawing strain. At the same time, compared with high-strength material, the damage growth rate of low-strength material is relatively smaller in the early stage of drawing, and the maximum damage after the first drawing pass is almost zero. In addition, the increase of die angle significantly increases the drawing damage. When the angle is $\alpha = 12^\circ$, after five drawing passes, the maximum damage values of high-strength material and low-strength material are 0.077 and 0.051, respectively. The maximum damage values of the two materials have been closer to their corresponding critical damage value of 0.15 (high-strength material) and 0.11 (low-strength material). According to the change law of drawing damage, if the drawing strain increases, steel wire will be seriously damaged. When the damage is large enough, the materials will crack directly from the maximum damage position, and finally lead to the failure of wire drawing.

Fig. 12 is the trend of the maximum section stress changing with the drawing strain when $\alpha = 6^\circ$, where the absolute values of positive and negative section stresses of each drawing pass are almost similar no matter it is high-strength material or low-strength material. With the increase of drawing strain, the stress and stress growth rate also increase. By comparing the ordinate range of Fig. 12 (A) and (B), it is obvious that under each drawing strain, the section stress formed by high-strength material is greater than that of low-strength material. The maximum absolute values of section stress produced by the two kinds of materials is largely different, which is 533.4 MPa and 362.3 MPa, respectively. In the process of multi-pass drawing, the continuous increase of section stress leads to the increase of compressive stress of wire after drawing, which also increases the fracture possibility of wire in multi-pass drawing.

To sum up, Fig. 11 and Fig. 12 demonstrate that there is a significant difference in the maximum damage of materials with different strength during drawing. With the increase of drawing strain, the damage degree of the two materials will gradually increase. And the maximum damage of high-strength material during drawing is obviously higher than that of low-strength material. Besides, the section stress on the outer surface of wire also varies greatly, and high-strength material have higher stress. Based on the above analysis, compared with lower-strength material, high-strength material has a greater risk of splitting fracture.

5.2. Effect of die half-angle on splitting

Fig. 13 is the trend of the maximum damage value of the same strength materials at different die half-angles. The figure demonstrates that the maximum damage of both material is related to die angle. The larger the angle is, the greater the corresponding drawing damage to each pass is. This result has a certain correlation with the conclusion in the reference [46] that the equivalent plastic strain increases with the increase of the die angle. This also shows that the die angle is a key factor in drawing damage. Besides, the growth rate of damage at all angles decreases as the drawing strain increases. With the increase of die angle, the maximum damage interval among each pass progressively decreases whereas the damage growth rate continues to increase.

In order to understand the influence of die angle on section stress of wire, Fig. 14 calculates the absolute value of maximum section stress after the first pass drawing at each angle. It is found that when material is the same, the absolute value of maximum stress fluctuates with the change of die angle, but the fluctuation value is small. At the same time, the absolute value of stress tends to be horizontally linear after fitting, and its growth rate is also close to zero. The absolute values of positive and negative section stresses are almost the same in different angles.

According to the analysis of Fig. 13 and Fig. 14, with the increase of die angle, the core damage increases accordingly, but the section stress changes little. Therefore, it can be explained that the increase of die angle also increases the failure probability of wire drawing.

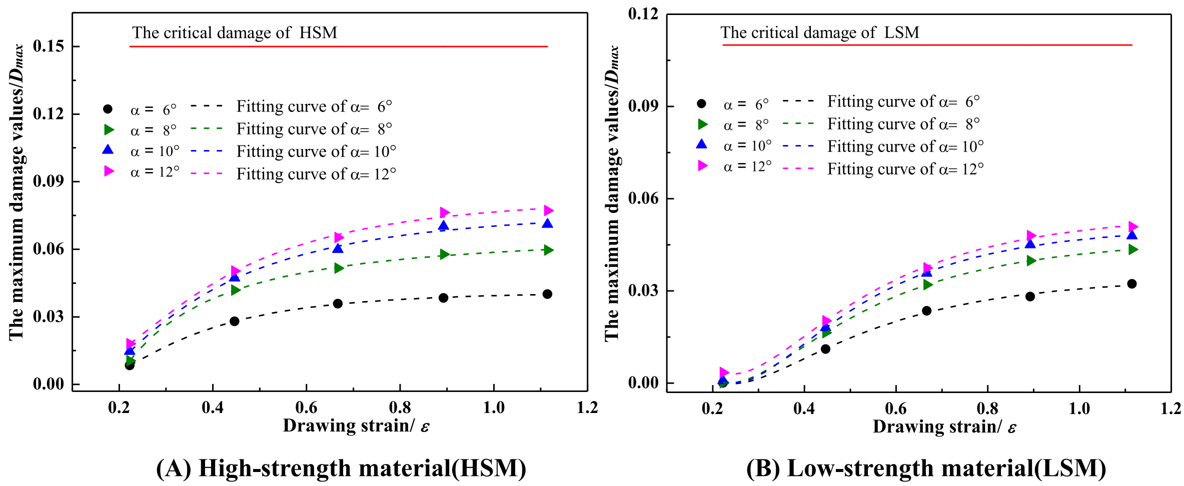


Fig. 13. The drawing damage of materials with different strength.

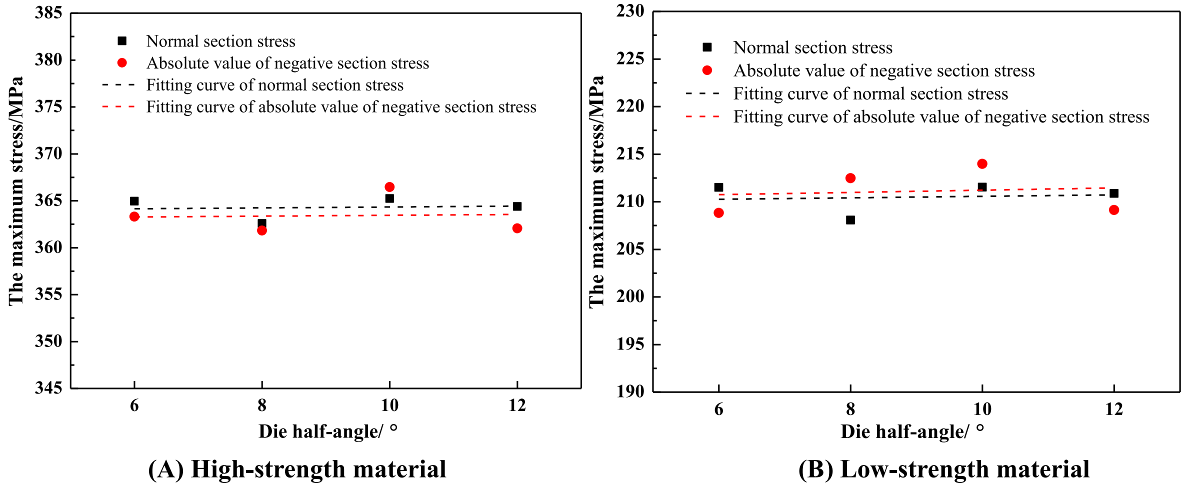


Fig. 14. Section stress tendency of different die half-angles (after 1st pass).

6. Conclusions

Aiming at splitting fracture in the process of multi-pass drawing, the crack morphology and microstructure of wire are analyzed in the paper by means of SEM. In the meantime, the formation mechanism of splitting fracture is researched via numerical simulation. Based on the analysis of experiments and simulation results, the major conclusions can be briefly summarized as follows:

(1) The experiment revealed that splitting fracture is formed by the crack passing through the core and the outer surface. Numerical simulation illustrates that this process is caused by the combination of core damage and near-surface symmetrical compression stress. The wire core becomes the source of crack due to its serious damage, and then the crack propagates under the action of symmetrical compression stress, and splitting fracture is thus formed.

(2) The drawing damage increases logarithmically with the increase of drawing strain under the same die angle. Meanwhile, the increase of the angle will also enhance the damage value. In addition, the drawing damage of high-strength material is significantly higher than that of low-strength material under the same drawing condition.

(3) After each drawing pass, a kind of symmetrical compression stress is distributed on the near surface of wire, and the stress increases as the drawing strain increases. Under the same drawing strain, the stress changes little at all angles. Furthermore, the surface stress of high-strength material is significantly higher than that of low-strength material.

(4) Through the overall comparison and analysis for the core damage and external compressive stress produced in the drawing process, it can be judged that the drawing of low-strength materials at a lower die angle will be beneficial to reduce the wire broken rate and improve drawing production efficiency. This research has a certain reference value for the improvement of industrial drawing process.

CRediT authorship contribution statement

Ao Ma: Writing – original draft, Methodology, Software, Visualization, Validation, Formal analysis. **Jiaxing Cheng:** Validation, Visualization, Supervision, Writing – review & editing. **Dasheng Wei:** Data curation, Visualization, Validation. **Qiang Li:** Data curation, Visualization. **Feng Fang:** Supervision, Resources, Writing – review & editing. **Zhaoxia Li:** Conceptualization, Supervision, Methodology, Project administration, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The research presented in this paper was supported by Industry-University Research Cooperation Project of Jiangsu Province, China (BY2018194) and Baosteel Golden Apple Project. The research was also supported by the National Natural Science Foundation of China (Grant No. 52171110).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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